AJAY GANESAN IYER

Motion Planning | State Estimation | Robotics & Controls

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EDUCATION

Texas A&M University, College Station, TX Aug 2024 - Present
Ph.D., Mechanical Engineering

Indian Institute of Technology, Roorkee, India Jun 2021
M.Tech., Systems and Control

National Institute of Technology, Tiruchirappalli, India May 2019
B.Tech., Instrumentation and Control

RESEARCH EXPERIENCE

Research Assistant, Texas A&M University **Aug 2026 - Present**
Frontier Mapping | Advisor: Dr. Srikanth Saripalli

- Developing **vision-language model (VLM)-based semantic value mapping for frontier exploration** to guide further inspection when clutter, occlusion, and limited viewpoints make geometric coverage insufficient in unstructured environments.
- Formulating a **2D semantic value map** by projecting image-text relevance scores using depth and robot pose and fusing observations with visibility-based weights. Developing frontier ranking from map values, expected information gain, and travel cost to select inspection viewpoints.
- Building a **ROS 2/Gazebo Husky-Jackal framework** with LiDAR mapping, cameras, coordinate transforms, and extended Kalman filter localization to evaluate the proposed frontier selection under occlusion and restricted viewpoint access.

Graduate Research Assistant, Texas A&M University **Aug 2025 - Jul 2026**
Autonomous Vehicles for All | Advisor: Dr. Reza Langari

- Developed a **Frenet-frame motion planner** for marked and unmarked gravel roads, generating candidate trajectories from lateral offsets and velocity profiles and selecting feasible paths using curvature constraints and a weighted cost function.
- Supplied the planner with **road-boundary estimates from a real-time ROS perception pipeline**. Tuned CLRerNet lane detection and SAM drivable-area segmentation and aligned camera, LiDAR, and odometry data in a common map frame.
- Tested the integrated pipeline using **ROS bags and simulation** under low light, glare, dust, unmarked gravel, sensor dropout, and timing jitter. Addressed degraded trajectory candidates through perception-threshold tuning and temporal smoothing to maintain lane continuity.
- Studied **visual-token redundancy in OpenDriveVLA** to identify opportunities for efficient Vision-Language-Action (VLA) inference. Evaluated principal component analysis and token selection across **6,019 nuScenes validation samples** using UniAD and STP-3 planning metrics.
- Identified a compact representation of the visual inputs: approximately **10 principal components** captured **80% of scene-token variance** and **97% of map-token variance**, providing a basis for inference-time token reduction.

Graduate Research Assistant, Texas A&M University **Aug 2024 - Aug 2025**
Robotic Welding Automation | Advisor: Dr. Prabhakar Pagilla

- Built a **Python/PyQt5/ROS interface** to automate **UR10e/Miller pulsed-MIG** weld setup and execution, integrating Arduino PWM/relay control for arc voltage, wire feed, and weld triggering.
- Registered CAD parts using **three torch-probed reference points, Kabsch alignment, and ICP refinement**. Generated STL-edge weld paths with surface-normal orientations and a 45° torch tilt without external metrology.
- Implemented **Newton-Raphson IK** and **RRT*** with wrist-limit handling and angle unwrapping. **Time-parameterized trajectories** for constant-tip-speed welding and coordinated weld triggering with approach/retreat motions. Added trajectory logging and joint-jump/jerk checks.
- Added **Optris PI 05M infrared weld-pool monitoring** to the execution pipeline, using bead-height and calibrated temperature feedback for closed-loop regulation of travel speed and arc voltage.

TECHNICAL SKILLS

Software: Python, C++, C, Embedded C, MATLAB/Simulink, ROS 1/2, Git, Gazebo, RViz, CARLA, Webots
Planning & Control: Model Predictive Control (MPC), constrained optimization, RRT*, inverse kinematics, sliding mode control, PID, system identification, stability testing
Estimation & Perception: Kalman filtering, camera/LiDAR fusion, OpenCV, semantic segmentation, PCA, TensorFlow, Keras, Pandas
Hardware & Interfaces: UR10e, Arduino, Raspberry Pi, LabVIEW, PyQt5, infrared thermography
Coursework: Aerial Robotics, Control of Mobile Robots, Computational Motion Planning, Optimal Control, Spacecraft Dynamics and Control, Deep Learning

INDUSTRY EXPERIENCE

Robotics Engineer, MIKO **Jun 2021 - Jun 2023**

Mumbai, India

- Developed **control and path-planning software for a head-motion manipulator**, using system identification, stability testing, and robust controller synthesis to produce expressive joint-level articulation.
- Extended the control work to a **self-balancing robot**, integrating the sensor suite and tuning robust controllers to maintain balance.
- Connected **audio direction-of-arrival estimates to yaw-axis control** so the robot could turn toward an active speaker in real time. Also delivered remote-control functionality for the mobile platform.

RELATED PROJECTS

LPV Model Predictive Control for Autonomous Vehicles **Sep 2024 - Nov 2024**

Texas A&M University

- Developed **constrained model predictive control in Python** for vehicle trajectory tracking, using a six-state nonlinear bicycle model with steering and acceleration inputs and a **20 ms discretization step**.
- Updated the linearized dynamics along the prediction horizon and formulated a **10-step quadratic program solved with cvxopt**, combining tracking error, terminal error, and input-rate penalties through an augmented input-increment model.
- Constrained the optimization by **steering, speed, sideslip, and tire-force limits** ($\pm 30^\circ$ steering, 1-30 m/s longitudinal speed). Validated the resulting closed-loop controller on straight, sinusoidal, and circuit trajectories.

Robust Nonlinear Control for Urban Vehicle Platooning **Aug 2020 - Apr 2021**

M.Tech. Thesis, IIT Roorkee

- Designed longitudinal and lateral controllers for a **three-vehicle platoon** to maintain spacing and track a reference path at 8 m/s, using nonlinear bicycle dynamics in MATLAB/Simulink.
- Implemented and compared **PID, conventional sliding mode, power-rate exponential reaching law (PREL), and super-twisting control**, with a **Luenberger observer** to estimate yaw rate from lateral velocity.
- Assessed the controllers using tracking-error integrals and control energy across all three vehicles. **PREL achieved the lowest error indices**, while super-twisting provided lateral convergence without overshoot and the **lowest longitudinal control energy**.

Kalman Filtering for Needle-Tip State Estimation **Sep 2019 - Dec 2019**

Indian Institute of Technology, Roorkee

- Built a **LabVIEW needle-tip tracker** combining background subtraction, edge detection, ROI-based RANSAC line fitting, and a **four-state Kalman filter** to maintain position estimates when ultrasound speckle obscured the tip.
- Assessed tracking accuracy against an **NDI-Aurora electromagnetic tracker** in artificial and biological tissue at insertion depths up to 6.5 cm, using position root mean square error (RMSE).
- Compared with vision-only tracking, Kalman filtering reduced needle-tip position RMSE **from 0.79 cm to 0.03 cm in artificial tissue** and **from 0.84 cm to 0.21 cm in biological tissue**.

RELEVANT INTERNSHIP

Indian Institute of Technology, Bombay | EKF-SLAM **Jun 2017 - Jul 2017**

Mobile Robot Localization, Mapping and Navigation

- Adapted a MATLAB **Extended Kalman Filter SLAM (EKF-SLAM)** simulator for joint robot-pose and landmark estimation under motion and sensor noise, using Jacobian linearization and Mahalanobis-gated nearest-neighbor data association.
- Observed reduced landmark uncertainty during repeated waypoint traversals. Extended the simulator with **reactive steering from three sensor views** to navigate around landmarks without preset waypoints.
- Implemented navigation on a physical **SPARK-V robot in Embedded C (ATmega16)**, combining ultrasonic obstacle avoidance with wheel-encoder distance feedback. Demonstrated collision-free traversal of a room containing obstacles.

PUBLICATIONS & AWARDS

- A. G. Iyer, J. Samantaray, S. Ghosh, A. Dey, S. Chakrabarty. "Sliding Mode Control Using Power Rate Exponential Reaching Law for Urban Platooning." *7th International Conference on Advances in Control and Optimization of Dynamical Systems (ACODS)*, 2022.
- M. Adharsh, Ajay G. Iyer, Abhinav G. Athrey, Ashwin V., Venkatesh V. G. "Particle Swarm Optimization Algorithm for Finding the Brightest Spot in an Arena." *3rd IEEE International Conference on Electronics, Communication and Aerospace Technology*, 2019. **Best Paper Award**, 2019.